Probabilistic Machine Learning

Chapter 6: Information Theory

Slides based on K. Murphy, *Probabilistic Machine Learning: An Introduction*

Intro to ML
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Outline

1. Entropy
2. ⋆ Relative entropy (KL divergence)
3. ⋆ Mutual information

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Entropy as a measure of uncertainty

- Entropy measures the uncertainty, or lack of predictability, of a random variable's distribution.
- Also interpretable as the **information content** of a data source: high-entropy $p \Rightarrow$ observations $X_n \sim p$ are hard to predict $\Rightarrow$ the dataset carries a lot of information.
- A degenerate distribution (entropy 0) always produces the same value $\Rightarrow$ no information.

Intuition

Entropy tells you, on average, how many yes/no questions you need to pin down the outcome of a random draw.

Entropy of a discrete random variable

Key equation

$$H(X) \triangleq -\sum_{k=1}^{K} p(X{=}k) \log_2 p(X{=}k) = -\mathbb{E}_X[\log p(X)]$$

- Base-2 log $\Rightarrow$ units of **bits**; base-e log $\Rightarrow$ **nats**.
- Example: $p = [0.25, 0.25, 0.2, 0.15, 0.15]$ over 5 states $\Rightarrow H = 2.29$ bits.
- **Maximum entropy**: the uniform distribution, $p(x{=}k) = 1/K \Rightarrow H(X) = \log_2 K$.
- **Minimum entropy** (0): any delta function — no uncertainty at all.

Binary entropy function

$X \in \{0, 1\}$, $p(X{=}1) = \theta$:

$$H(\theta) = -[\theta \log_2 \theta + (1 - \theta) \log_2 (1 - \theta)]$$

- Maximum value 1 bit at $\theta = 0.5$ (fair coin — needs exactly one yes/no question).
- $H(\theta) \to 0$ as $\theta \to 0$ or $\theta \to 1$: an (almost) deterministic outcome carries (almost) no information.

Application: DNA sequence logos

- Align DNA sequences; estimate the empirical distribution $\hat{\theta}_t$ over $\{A, C, G, T\}$ at each position t — a **position weight matrix** / sequence motif.
- **Sequence logo**: scale column t by $R_t = 2 - H_t$ (max possible entropy over 4 letters is $\log_2 4 = 2$).
- Deterministic (entropy-0) positions get maximal height 2 — these are highly conserved, often functionally important, sites.
- The **consensus sequence** just takes the most likely letter at each position, ignoring uncertainty.

Estimating entropy is hard for large alphabets

- Estimating $H(X)$ requires estimating $p(X)$, which needs a lot of data when $|\mathcal{X}|$ is large.
- Example: X = identity of a word in an English document — long tail of rare words, new words constantly invented $\Rightarrow$ $p(X)$ (and hence $H(X)$) is hard to pin down reliably.

Cross entropy

Key equation

$$H_{ce}(p, q) \triangleq -\sum_{k=1}^{K} p_k \log q_k$$

- Interpretation: expected number of bits to compress data from p using a code built for q.
- Minimized by $q = p$, giving $H_{ce}(p, p) = H(p)$ — **Shannon's source coding theorem**: you cannot compress below the entropy.

Joint entropy

$$H(X, Y) = -\sum_{x,y} p(x, y) \log_2 p(x, y)$$

Example: Even / prime numbers, $n \in \{1, \ldots, 8\}$

$X(n)$=1 if n even, $Y(n)$=1 if n prime. Marginals both uniform: $H(X) = H(Y) = 1$ bit. Joint distribution has probabilities $1/8, 3/8, 3/8, 1/8$, giving $H(X, Y) = 1.81$ bits $< H(X) + H(Y) = 2$ bits.

$$\max\{H(X), H(Y)\} \leq H(X, Y) \leq H(X) + H(Y)$$

Upper bound tight iff $X \perp Y$; correlation reduces the joint entropy below the sum.

Conditional entropy and the chain rule

Key equation

$$H(Y \mid X) \triangleq \mathbb{E}_{p(X)}\big[H(p(Y \mid X))\big] = H(X, Y) - H(X)$$

- Y deterministic function of $X \Rightarrow H(Y \mid X) = 0$; $X \perp Y \Rightarrow H(Y \mid X) = H(Y)$.
- In general $H(Y \mid X) \leq H(Y)$: **on average**, conditioning on data never increases uncertainty (though for a specific x it might).

Key equation

$$H(X_1, \ldots, X_n) = \sum_{i=1}^{n} H(X_i \mid X_1, \ldots, X_{i-1}) \quad \text{(chain rule for entropy)}$$

Worked example: chain rule for entropy

Consider three variables X_1, X_2, X_3 (e.g., successive words in a sentence). The chain rule lets us decompose the joint entropy sequentially:

$$H(X_1, X_2, X_3) = H(X_1) + H(X_2 \mid X_1) + H(X_3 \mid X_1, X_2)$$

- This is exactly how autoregressive sequence models (Ch. 15) factorize a joint distribution: $p(x_{1:T}) = \prod_t p(x_t \mid x_{<t})$, and the cross-entropy training loss is a Monte Carlo estimate of $\sum_t H(X_t \mid X_{<t})$.
- If the X_i are iid, this collapses to $H(X_1, X_2, X_3) = 3H(X_1)$.

Perplexity

$$\text{perplexity}(p) \triangleq 2^{H(p)}$$

- Uniform p over K states $\Rightarrow$ perplexity $= K$; minimum perplexity is 1 (perfectly predictable).
- For evaluating a language model q against empirical data distribution p_D: $\text{perplexity}(p_D, q) = 2^{H_{ce}(p_D, q)}$ — the "exponentiated cross entropy", geometric mean of inverse predicted probabilities.
- Interpretable as the **weighted average branching factor** — effective number of equally-likely next tokens.
- Since $H(p^*) \leq H_{ce}(p^*, q)$, perplexity can never go below the true entropy of the data-generating process.

⋆ Differential entropy (continuous rvs)

⋆ Advanced / optional

$$h(X) \triangleq -\int_{\mathcal{X}} p(x) \log p(x)\, dx$$

- Example: $X \sim \mathrm{Unif}(0, a) \Rightarrow h(X) = \log a$.
- Unlike discrete entropy, **differential entropy can be negative**: e.g., $X \sim \mathrm{Unif}(0, \frac{1}{8})$ gives $h(X) = -3$.
- Interpretation: entropy of an n-bit quantization of X is $\approx h(X) + n$.

⋆ Entropy of a Gaussian

⋆ Advanced / optional

Key equation

$$h(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})) = \tfrac{1}{2} \ln |2\pi e \boldsymbol{\Sigma}| = \tfrac{d}{2} + \tfrac{d}{2} \ln(2\pi) + \tfrac{1}{2} \ln |\boldsymbol{\Sigma}|$$

1D case: $h(\mathcal{N}(\mu, \sigma^2)) = \frac{1}{2} \ln(2\pi e \sigma^2)$.

Entropy vs. variance: entropy grows monotonically with σ^2 for a single Gaussian, but *not* in general — a bimodal mixture with modes pushed far apart has ever-growing variance but roughly constant entropy (still uncertain which mode a sample came from).

⋆ Discretizing to estimate differential entropy

⋆ Advanced / optional

Computing $h(X)$ directly can be hard; a common approximation bins the data using empirical quantiles. Scott's rule for the number of bins:

$$B = N^{1/3} \frac{\max(D) - \min(D)}{3.5\,\sigma(D)}$$

Does not scale well to multivariate X (curse of dimensionality).

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⋆ KL divergence: definition

⋆ Advanced / optional

A **divergence** $D(p, q) \geq 0$ with equality iff $p = q$ (need not be symmetric or satisfy the triangle inequality, unlike a metric).

Key equation

$$D_{\mathrm{KL}}(p\|q) \triangleq \sum_k p_k \log \frac{p_k}{q_k} \quad \text{or} \quad D_{\mathrm{KL}}(p\|q) \triangleq \int p(x) \log \frac{p(x)}{q(x)} \, dx$$

⋆ KL divergence: interpretation

⋆ Advanced / optional

$$D_{\mathrm{KL}}(p\|q) = \underbrace{\sum_k p_k \log p_k}_{-H(p)} - \underbrace{\sum_k p_k \log q_k}_{-H_{ce}(p,q)\text{ term}} = H_{ce}(p,q) - H(p)$$

Since $H_{ce}(p,q)$ lower-bounds the bits needed to compress p-data with a q-based code, $D_{\mathrm{KL}}(p\|q)$ is the **extra number of bits** you pay for using the wrong distribution q instead of the true p.

⋆ Example: KL between two Gaussians

⋆ Advanced / optional

$$D_{\mathrm{KL}}(\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) \| \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)) = \frac{1}{2}\left[\mathrm{tr}(\boldsymbol{\Sigma}_2^{-1}\boldsymbol{\Sigma}_1) + (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}_2^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) - D + \log \frac{\det \boldsymbol{\Sigma}_2}{\det \boldsymbol{\Sigma}_1}\right]$$

Scalar case:

$$D_{\mathrm{KL}}(\mathcal{N}(\mu_1, \sigma_1^2) \| \mathcal{N}(\mu_2, \sigma_2^2)) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

⋆ Non-negativity of KL (information inequality)

⋆ Advanced / optional

Theorem: Jensen's inequality (convex f)

$f(\sum_i \lambda_i x_i) \le \sum_i \lambda_i f(x_i)$, $\lambda_i \ge 0, \sum_i \lambda_i = 1$. For concave log: $\log \mathbb{E}[g(x)] \ge \mathbb{E}[\log g(x)]$.

Theorem: Information inequality

$D_{\mathrm{KL}}(p\|q) \ge 0$, with equality iff $p = q$.

Proof sketch: apply Jensen's inequality (concavity of log) to $-D_{\mathrm{KL}}(p\|q) = \sum_{x\in A} p(x) \log \frac{q(x)}{p(x)} \le \log \sum_{x\in A} q(x) \le \log \sum_x q(x) = \log 1 = 0$, where $A = \{x : p(x) > 0\}$.

⋆ Corollary: uniform maximizes entropy

⋆ Advanced / optional

Theorem:

$H(X) \leq \log |\mathcal{X}|$, with equality iff $p(x)$ is uniform.

Proof: let $u(x) = 1/|\mathcal{X}|$. Then

$$0 \leq D_{\mathrm{KL}}(p\|u) = \sum_x p(x) \log \frac{p(x)}{u(x)} = \log |\mathcal{X}| - H(X)$$

⋆ KL divergence and MLE

⋆ Advanced / optional

Minimize $D_{\mathrm{KL}}(p\|q)$ over q; let $p = p_D$ be the empirical distribution, $p_D(x) = \frac{1}{N}\sum_n \delta(x - x_n)$:

$$D_{\mathrm{KL}}(p_D\|q) = -\frac{1}{N}\sum_n \log q(x_n) + C$$

where C doesn't depend on q. This is exactly the **average NLL** — minimizing KL to the empirical distribution $\equiv$ **maximum likelihood estimation**.

Intuition

This exposes a flaw of plain MLE: it treats the empirical distribution (finite spikes on observed points) as if it were the true distribution — motivating data augmentation and smoothing.

⋆ Forwards (inclusive) KL: mode-covering

⋆ Advanced / optional

Approximate p with a simpler q by minimizing

$$D_{\mathrm{KL}}(p\|q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx \qquad (\text{“M-projection”})$$

Wherever $p(x) > 0$ but $q(x) = 0$, the integrand blows up. Minimizing therefore forces q to put mass *everywhere* p does: q is **zero-avoiding** / **mode-covering**, and typically *over*-estimates the support of p (e.g., a unimodal q straddling both modes of a bimodal p, landing mass in the low-density valley between them).

⋆ Reverse (exclusive) KL: mode-seeking

⋆ Advanced / optional

$$D_{\mathrm{KL}}(q\|p) = \int q(x) \log \frac{q(x)}{p(x)}\, dx \qquad (\text{“I-projection”})$$

Wherever $p(x) = 0$ but $q(x) > 0$, the integrand blows up. Minimizing therefore forces q to avoid all zero-probability regions of p: q is **zero-forcing** / **mode-seeking** — it typically locks onto a single mode and *under*-estimates the support of p.

- This is the objective used by **variational inference** (approximate posteriors), which is why VI posteriors are often overconfident / too concentrated.

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⋆ Mutual information: definition

⋆ Advanced / optional

Key equation

$$I(X;Y) \triangleq D_{\mathrm{KL}}\big(p(x,y) \,\|\, p(x)p(y)\big) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

Non-negative (it's a KL divergence); $I(X;Y) = 0$ iff $X \perp Y$.

⋆ Mutual information: interpretation

⋆ Advanced / optional

MI = information gain from replacing the independence model $p(x)p(y)$ with the true joint $p(x, y)$. Equivalently:

Key equation

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y)$$

Interpretation: the **reduction in uncertainty** about X from observing Y (or vice versa, by symmetry). Also gives an alternative proof that conditioning reduces entropy on average: $I(X;Y) \geq 0 \Rightarrow H(X \mid Y) \leq H(X)$.

⋆ Example: mutual information of even/prime

⋆ Advanced / optional

Continuing the even/prime example: $H(X) = H(Y) = 1$ bit, conditional distribution normalized per row gives $H(Y \mid X) = 0.81$ bits, so

$$I(X;Y) = H(Y) - H(Y \mid X) = 1 - 0.81 = 0.19 \text{ bits}$$

Check: $H(X,Y) = H(X \mid Y) + I(X;Y) + H(Y \mid X) = 0.81 + 0.19 + 0.81 = 1.81$ bits. ✓

⋆ Conditional mutual information & chain rule

⋆ Advanced / optional

$$I(X;Y \mid Z) \triangleq \mathbb{E}_{p(Z)}[I(X;Y) \mid Z] = H(X \mid Z) - H(X \mid Y,Z)$$

Interpretation: extra information X gives about Y, beyond what Z already tells us.

Key equation

$$I(Z_1,\ldots,Z_N;X) = \sum_{n=1}^{N} I(Z_n;X \mid Z_1,\ldots,Z_{n-1}) \quad \text{(chain rule for MI)}$$

⋆ MI as a "generalized correlation coefficient"

⋆ Advanced / optional

Jointly Gaussian (X, Y) with correlation ρ, common variance σ^2:

$$I(X;Y) = -\tfrac{1}{2}\log(1-\rho^2)$$

- $\rho = 0$: $I = 0$ (independent, as expected).
- $\rho = \pm 1$: $I = \infty$ (perfect determinism, $X = \pm Y$).

For general (non-Gaussian, scalar) variables: estimate via discretization/binning, or non-parametrically via K-NN distances (the **KSG estimator**).

⋆ Normalized MI and the maximal information coefficient

⋆ Advanced / optional

Since $0 \leq I(X;Y) \leq \min(H(X), H(Y))$, define

$$\mathrm{NMI}(X,Y) = \frac{I(X;Y)}{\min(H(X), H(Y))} \in [0,1]$$

$\mathrm{NMI} = 1$ iff one variable is a deterministic function of the other.

Maximal Information Coefficient (MIC): search over 2D grids G, $\mathrm{MIC}(X,Y) = \max_G \frac{I((X,Y)|_G)}{\log \|G\|}$. Captures non-linear and non-functional relationships, not just linear correlation — called "a correlation for the 21st century". Exhibits **equitability**: similar scores for equally-noisy relationships regardless of functional form.

⋆ MIC: equitability and related statistics

⋆ Advanced / optional

- **Equitability**: MIC gives similar scores to equally-noisy relationships regardless of their functional form (linear, non-linear, even non-functional/one-to-many) — unlike R^2, which only captures linear trends.
- **MICe**: a faster estimator requiring only 1D grid search, computable in $O(n)$ via dynamic programming.
- **TICe** (total information content): sums $M(k, l)$ over all grid sizes rather than taking the max — higher power to detect weak relationships in small samples, at the cost of equitability.
- Practical workflow: use TICe to screen many candidate variable pairs (e.g., thousands of pairwise relationships in a large dataset), then MICe to quantify the strength of the surviving candidates.

⋆ Data processing inequality

⋆ Advanced / optional

Theorem: Data processing inequality (DPI)

If $X \to Y \to Z$ is a Markov chain (i.e., $X \perp Z \mid Y$), then $I(X;Y) \geq I(X;Z)$.

Proof: chain rule for MI gives $I(X;Y,Z) = I(X;Z) + I(X;Y \mid Z) = I(X;Y) + I(X;Z \mid Y)$. Since $X \perp Z \mid Y$, $I(X;Z \mid Y) = 0$, so $I(X;Z) + I(X;Y \mid Z) = I(X;Y)$; since $I(X;Y \mid Z) \geq 0$, we get $I(X;Y) \geq I(X;Z)$.

Intuition

Processing data can only destroy information about X, never create it.

⋆ Sufficient statistics

⋆ Advanced / optional

From the DPI applied to $\theta \to D \to s(D)$: $I(\theta; s(D)) \leq I(\theta; D)$.

Definition: Sufficient statistic

If equality holds, $s(D)$ is a **sufficient statistic** for θ — it summarizes D without losing information relevant to θ.

Definition: Minimal sufficient statistic

A sufficient $s(D)$ such that every other sufficient statistic $s'(D)$ can be mapped onto it: $s(D) = f(s'(D))$ — maximal compression without information loss.

Examples: for N Bernoulli trials, $(N, \sum_n \mathbb{I}[x_n{=}1])$ is minimal sufficient; for a Gaussian with known variance, the sample mean and N suffice.

⋆ Fano's inequality

⋆ Advanced / optional

Bounds achievable classification error in terms of mutual information. Let $\hat{Y} = f(X)$ with $Y \to X \to \hat{Y}$ Markov, $P_e = P(Y \neq \hat{Y})$:

$$H(Y \mid X) \leq H(Y \mid \hat{Y}) \leq H(P_e) + P_e \log |\mathcal{Y}|$$

Since $H(P_e) \leq 1$:

$$P_e \geq \frac{H(Y \mid X) - 1}{\log |\mathcal{Y}|}$$

Intuition

Justifies feature selection by mutual information: minimizing $H(Y \mid X)$ (i.e., maximizing $I(X; Y)$) lowers the achievable error floor for *any* classifier built on features X.

⋆ Fano's inequality: proof sketch

⋆ Advanced / optional

Let $E = \mathbb{I}\left[\hat{Y} \neq Y\right]$. Chain rule for entropy on $(E, Y \mid \hat{Y})$, two ways:

$$H(E, Y \mid \hat{Y}) = \underbrace{H(Y \mid \hat{Y}) + H(E \mid Y, \hat{Y})}_{H(E \mid Y, \hat{Y}) = 0} = H(E \mid \hat{Y}) + H(Y \mid E, \hat{Y})$$

- Conditioning reduces entropy: $H(E \mid \hat{Y}) \leq H(E)$.
- Split on E: $H(Y \mid E, \hat{Y}) = (1 - P_e) \cdot 0 + P_e \cdot H(Y \mid \hat{Y}, E{=}1) \leq P_e \log |\mathcal{Y}|$.
- Combine: $H(Y \mid \hat{Y}) \leq H(E) + P_e \log |\mathcal{Y}|$.
- Finally apply the DPI ($I(Y; \hat{Y}) \leq I(Y; X) \Rightarrow H(Y \mid X) \leq H(Y \mid \hat{Y})$) to get the stated bound.

⋆ KL divergence $\approx \chi^2$ statistic for close distributions

⋆ Advanced / optional

When $p(x) \approx q(x)$, write $p(x) = q(x) + \Delta(x)$, so $p(x)/q(x) = 1 + \Delta(x)/q(x)$. Using $\log(1+\epsilon) \approx \epsilon - \epsilon^2/2$:

$$D_{\mathrm{KL}}(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \approx \frac{1}{2} \sum_x \frac{(p(x) - q(x))^2}{q(x)} = \frac{1}{2}\chi^2$$

Intuition

For nearby distributions, KL divergence behaves like (half) a χ^2 statistic — a quadratic, locally-Euclidean measure of discrepancy. This is why second-order (Fisher information) approximations to KL show up throughout the book (e.g., natural gradient, Laplace approximations, BIC in Ch. 5).

Plug-in entropy estimation: bias with finite data

- The naive **plug-in estimator** replaces p with the empirical distribution $\hat{p}$ and computes $H(\hat{p})$.
- This is **biased downward**: finite samples make the empirical distribution look "spikier" (lower entropy) than the true one, especially when $|\mathcal{X}|$ is large relative to N.
- Rule of thumb: reliable entropy estimation needs $N \gg |\mathcal{X}|$; with a long-tailed vocabulary (e.g., natural language) this is often violated, motivating smoothed/Bayesian entropy estimators.

Chapter summary

- **Entropy** $H(X)$ quantifies uncertainty/information content; maximized by the uniform distribution, minimized (0) by a deterministic outcome.
- **Cross-entropy**, **joint entropy**, **conditional entropy**, and the **chain rule for entropy** extend this to multiple variables and to comparing distributions.
- **Perplexity** 2^H gives an interpretable "effective branching factor", widely used to evaluate language models.
- **KL divergence*** measures the extra coding cost of using the wrong distribution; always ≥ 0; underlies MLE, and forward vs. reverse KL have very different (covering vs. seeking) behavior.
- **Mutual information*** $I(X;Y)$ is a symmetric, non-negative measure of statistical dependence; obeys a data processing inequality and underlies feature selection (Fano's inequality) and sufficient statistics.